



OMNIKON

NATIONAL HACKATHON 2026



ONE MISSION. BUILD THE IMPOSSIBLE.

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THEME:

Healthcare & Biomedical Technology

PROBLEM STATEMENT

Omni_BioTech_12 — Smartphone-Based Skin Cancer Screening

Dermatologists are scarce relative to population, delaying detection of skin conditions including cancers. Build an AI-assisted tool that helps flag suspicious skin lesions from smartphone images for further evaluation.

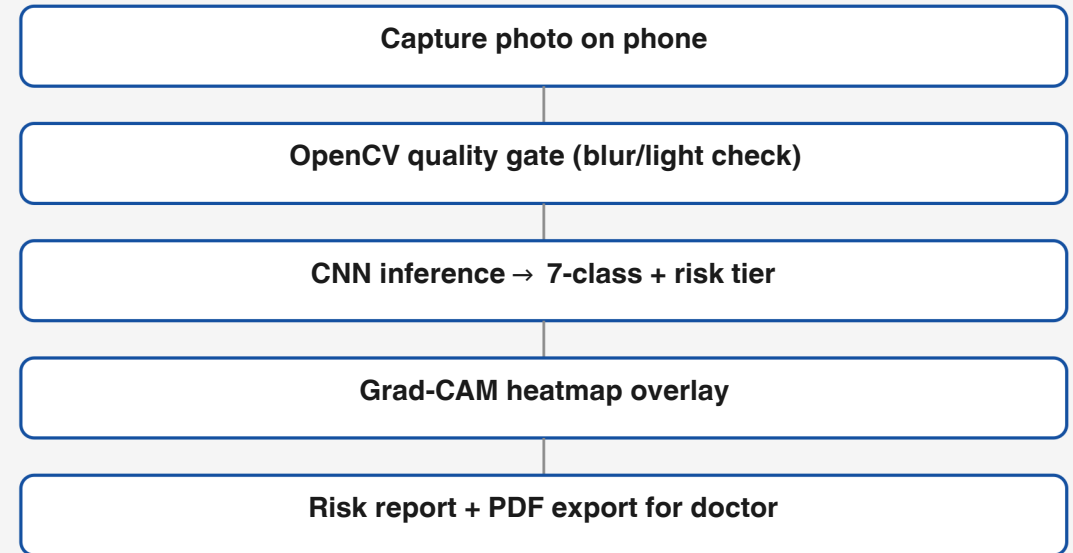
Proposed Solution

- **A mobile-first screening app: capture a lesion photo, run an on-the-spot AI risk check, and get a clear next step.**
 - Real-time image quality gate (blur / lighting check) runs before the AI, so bad photos never produce a false reading.
 - Lightweight CNN (EfficientNet-B0 / MobileNetV3) classifies the lesion into 7 standard dermoscopic categories.
- **Directly targets the core problem: it gives users in dermatologist-scarce areas an immediate, low-cost first opinion instead of a weeks-long wait.**
 - Risk-tiered output (Low / Moderate / High) replaces a scary raw score with an actionable recommendation.
- **Innovation: explainable AI + a built-in photo-quality gate — most student projects skip both and just show a bare accuracy number.**
 - Grad-CAM heatmaps let the user (and eventually a doctor) see exactly which part of the lesion drove the prediction — no black box.

Technologies Used

- **Mobile app:** Flutter / React Native (camera capture, results UI)
- **Backend:** FastAPI (Python), async REST endpoint
- **ML:** PyTorch, EfficientNet-B0 / MobileNetV3, Grad-CAM
- **Preprocessing:** OpenCV, Albumentations
- **Training data:** HAM10000, ISIC Archive (Kaggle / Colab free GPU)
- **Hosting:** Docker on Hugging Face Spaces / Render (free tier)

Methodology / Process Flow



Feasibility

- 100% buildable on free-tier tools: Colab/Kaggle GPU, HAM10000 dataset, Hugging Face Spaces hosting — no paid infrastructure required.
- Lightweight backbone (EfficientNet-B0) keeps training under a few GPU-hours and inference near real-time on a live demo.

Potential Challenges & Risks

- Class imbalance in HAM10000 (~67% benign) can bias the model toward always predicting “benign.”
- Regulatory / liability concerns if the tool is mistaken for a diagnostic device.
- Real-world smartphone photos vary widely in lighting and camera quality.

Mitigation Strategies

- Use Focal Loss + class weighting, and report Sensitivity/Recall on malignant classes, not just raw accuracy.
- Frame the tool strictly as an assistive triage aid, with a clear non-diagnostic disclaimer on every result.
- Enforce the OpenCV quality gate so poor-quality images are rejected before inference, rather than silently mis-scored.

Potential Impact on Target Audience

- Patients in rural/semi-urban areas get an immediate first read instead of waiting weeks or months for a dermatologist appointment.
- Encourages earlier action on suspicious lesions, when melanoma survival rates are highest.
- Gives clinicians a pre-screened, doctor-ready summary (photo + heatmap + risk tier), saving consultation time.

Benefits

- Social: wider access to a first-line screening tool, especially where dermatologists are scarce.
- Economic: reduces unnecessary specialist visits by triaging clearly low-risk cases, while flagging high-risk cases faster.
- Health outcomes: earlier detection directly improves survival odds for aggressive skin cancers like melanoma.

Details / Links of Reference and Research Work

- Dataset — HAM10000 (Human Against Machine, 10000 images), via Kaggle.
- Dataset — ISIC Archive (International Skin Imaging Collaboration).
- Reference tutorial — “Skin cancer lesion classification using the HAM10000 dataset,” DigitalSreeni (YouTube).
- Reference tutorial — “Classifying Skin Cancer,” Aladdin Persson (YouTube).
- Reference tutorial — Grad-CAM implementation walkthrough, The_CS_Hub (YouTube).
- Method reference — Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization (Selvaraju et al.).
- Model references — EfficientNet (Tan & Le) and MobileNetV3 (Howard et al.) architecture papers.